

GSM Signal Denoising and Modulation Classification for Rural Uganda

Final Year Project Presentation

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Rural mobile service is often present, but still too weak to depend on

Local service reality

- Basic GSM services still matter: voice, SMS, USSD, and Mobile Money.
- At the cell edge, a phone may show signal bars but still suffer noisy or unstable service.
- Weak I/Q structure makes reliable modulation interpretation difficult under low SNR.

Engineering interpretation

- The project treats weak service as a signal-quality problem, not only a coverage-map problem.
- Denoising is placed before classification so the AMC stage sees a cleaner signal representation.
- The same logic can support troubleshooting, spectrum monitoring, and rural service planning.

GSM-family I/Q

Working signal model

Baseband windows test noisy edge-of-coverage conditions.

DAE + AMC

Core technical pipeline

Denoising front end followed by automatic modulation classification.

Rural use case

Project motivation

Weak-signal interpretation can support practical network decisions.

Problem statement

Rural GSM users may experience unreliable service even where a network signal is visible. At the edge of coverage, received I/Q samples are degraded by low SNR, interference, fading, and unstable propagation. Conventional AMC methods can misclassify such noisy signals, making it harder to interpret the real service condition.

Technical gap

- Classical signal-processing filters are useful, but are usually fixed to known noise assumptions.
- Deep-learning AMC can classify raw I/Q samples, but low-SNR interference still reduces reliability.
- A denoising front end may recover useful signal structure before classification.

Project question

- Can denoising improve modulation classification under controlled low-SNR GSM-family conditions?
- Which SNR ranges benefit most from denoising-assisted classification?
- How should the results be interpreted for rural weak-service analysis?

Aim and objectives

Main objective: design and evaluate a denoising-assisted automatic modulation classification pipeline for noisy GSM-family I/Q signals in rural Uganda weak-service scenarios.

Specific objectives

- Define GSM-family low-SNR and interference scenarios for controlled evaluation.
- Design a denoising front end that reconstructs cleaner I/Q representations.
- Train and evaluate an AMC stage on raw noisy and denoised signal features.
- Compare baseline and denoising-assisted performance using accuracy, macro F1, confusion behavior, and signal plots.
- Interpret the findings for weak-service troubleshooting and rural connectivity planning.

Why this project matters

Justification

- Rural weak service is a practical problem for ordinary users and for network planning.
- Denoising before classification gives a measurable way to test whether useful signal structure can be recovered.
- The method is software-side and can be evaluated without transmitting or interfering with licensed networks.

Significance

- Supports better interpretation of noisy edge-of-coverage signals.
- Gives supervisors and reviewers visible before/after signal evidence.
- Creates a foundation for future Android/SDR receive-only prototype work.

Scope

- Controlled simulation and offline I/Q evaluation.
- GSM-family/proxy modulation classes used for AMC validation.
- No claim of completed field deployment or live service improvement.

Receive-only

Safety boundary

The project does not transmit into licensed spectrum.

Offline first

Evaluation boundary

The current results are controlled simulation outputs.

Prototype path

Next step

Android/SDR demonstration remains a future validation pathway.

Literature review: where the gap sits

Classical AMC

- Uses likelihood rules, moments, cyclostationary features, and engineered descriptors.
- Can be interpretable, but becomes fragile when assumptions about noise and channel conditions break.
- Useful as a baseline for explaining what the ML pipeline improves.

Deep-learning AMC

- Learns discriminative patterns directly from raw or transformed I/Q samples.
- Modern CNN and sequence models perform well when trained on representative SNR conditions.
- Performance still drops when signal structure is buried under low-SNR and interference.

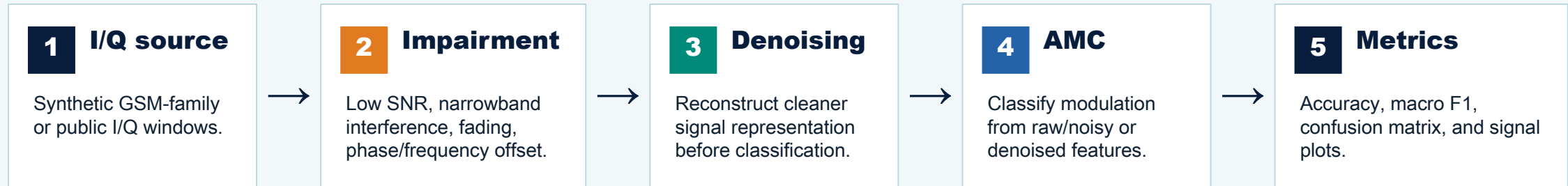
Denoising front ends

- Denoising autoencoders and related methods reconstruct cleaner signals before downstream tasks.
- The key research question is not only whether the signal looks cleaner, but whether classification improves.
- This project evaluates denoising as a preprocessing block for AMC under weak-service conditions.

Gap: few student-level implementations connect denoising, AMC metrics, and rural GSM weak-service interpretation in one reproducible pipeline.

Sources: ProjectReport.md literature review; DeepSig/RadioML datasets; low-SNR AMC and denoising references captured in reports/.

Methodology: denoising-assisted AMC signal chain



Evaluation comparison

- Baseline path: noisy I/Q features -> AMC classifier.
- Denoising-assisted path: impaired I/Q -> denoising front end -> AMC classifier.
- Both paths are evaluated under the same SNR levels and classes.
- This makes the difference attributable to the denoising front end.

Preliminary controlled simulation setup

4

Modulation classes

GMSK-like, QPSK, 8PSK, and 16QAM.

4

SNR levels

-8, -4, 0, and 4 dB controlled impairment settings.

1,920

Test windows

240 held-out windows per SNR and pipeline comparison.

20260524

Random seed

The run is deterministic and reproducible.

What was simulated

- Baseband I/Q windows with GSM-family/proxy modulation structure.
- Low-SNR AWGN, narrowband interference, mild fading, and phase/frequency offset.
- A denoising-assisted path compared against a noisy baseline path.

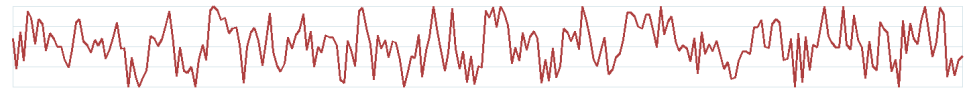
What was not claimed

- No field capture result is claimed in this slide deck.
- No live call, SMS, USSD, or Mobile Money improvement is claimed.
- No physical tower range extension is claimed.

Preliminary result: denoising makes weak signal structure easier to inspect

Preliminary controlled simulation: noisy vs denoised I/Q

Noisy received I component



Denoised I component



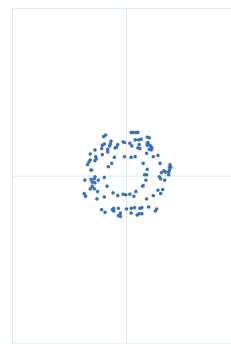
Clean reference I component



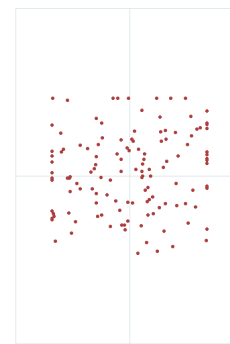
Noisy, denoised, and clean reference waveform comparison

Constellation comparison under low-SNR impairment

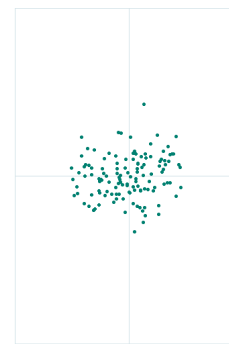
Clean reference



Noisy received



Denoised output

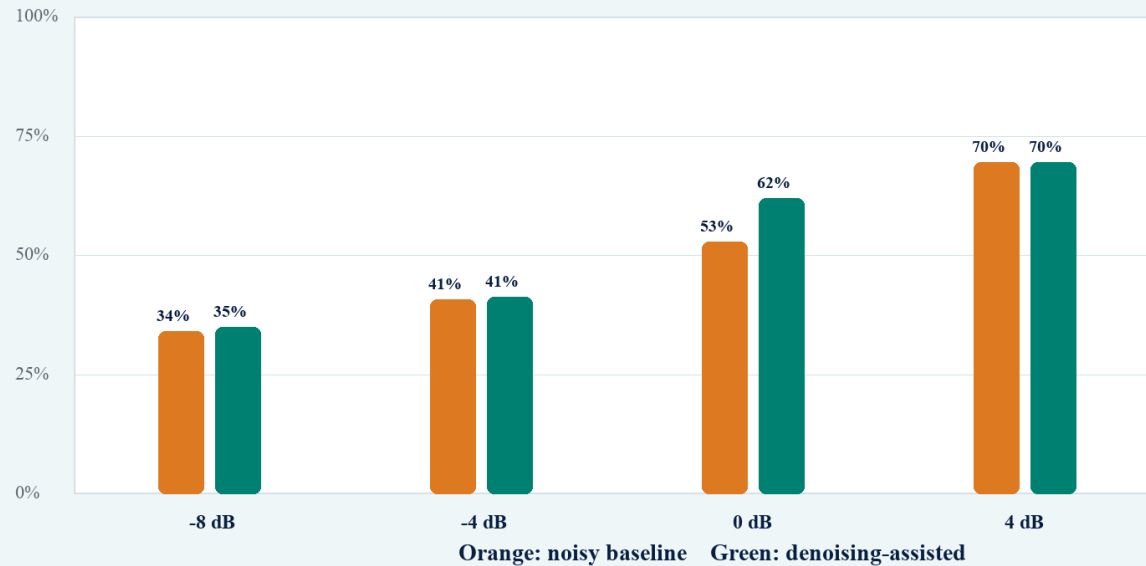


Constellation comparison at a low-SNR operating point

Interpretation: the preliminary denoising stage is not presented as final DAE performance. It is a controlled signal-processing validation layer that shows why denoising before AMC is worth completing.

Preliminary result: denoising-assisted AMC improves low-SNR classification

Preliminary AMC accuracy by SNR



Accuracy by SNR: noisy baseline versus denoising-assisted AMC

49.4%

Noisy baseline

Mean accuracy across controlled SNR levels.

52.0%

Denoising-assisted

Mean accuracy with the denoising-assisted path.

+9.2 pts

Largest observed gain

At 0 dB in this preliminary controlled run.

Result status: preliminary controlled simulation, not final field validation.

Discussion: what the preliminary results mean

Denoising-assisted confusion matrix

	GMSK-like	QPSK	8PSK	16QAM
GMSK-like	174	19	17	30
QPSK	11	73	91	65
8PSK	9	70	105	56
16QAM	4	29	60	147
True class				
	Predicted class			

Denoising-assisted confusion matrix across controlled SNR levels

Observed pattern

- The denoising-assisted path improves the controlled run overall.
- The clearest gain appears around the difficult 0 dB condition.
- Cleaner SNR conditions do not require aggressive denoising, so adaptive bypass is sensible.

Limitations

- The samples are synthetic and should be replaced or supplemented with stronger datasets.
- The denoising block is preliminary; final DAE training remains the next technical step.
- The results do not prove live service improvement or physical coverage extension.

Conclusion and recommendations

Conclusion

- Weak rural GSM service can be studied through noisy I/Q signal interpretation, not only through coverage-map labels.
- The proposed DAE-AMC chain is technically coherent: denoise first, classify second, evaluate across SNR.
- Preliminary controlled simulation results support continuing the denoising-assisted AMC implementation.

Recommendations

- Complete the full trained DAE and AMC implementation with stronger dataset coverage.
- Report results by SNR band using accuracy, macro F1, confusion matrices, and signal-quality plots.
- Use receive-only validation only; do not transmit, decode private content, or claim live-service gains without evidence.

Next 4 weeks

Implementation focus

DAE training, AMC benchmarking, final report update, and presentation rehearsal.

Final output

Expected deliverable

A reproducible denoising-assisted AMC evaluation for weak GSM-family signals.

Safety

Operating boundary

Receive-only, offline-first validation unless formal approval changes the scope.

References and artifact provenance

Selected references

- Uganda Communications Commission reports and service-quality materials referenced in ProjectReport.md.
- DeepSig / RadioML I/Q dataset literature for AMC evaluation framing.
- Low-SNR AMC and denoising-autoencoder papers summarized in reports/LowSNR_AMC_Denoising_Research.md.
- Project report manuscript: ProjectReport.md, final-year project workspace.

Preliminary simulation provenance

- Script: reports/preliminary_simulation_2026_05_24/run_preliminary_dae_amc.py
- Metrics: reports/preliminary_simulation_2026_05_24/results.csv
- Figures: signal_comparison.png, constellation_comparison.png, accuracy_by_snr.png, confusion_denoised.png
- Status: controlled simulation artifacts generated for presentation submission, not field deployment evidence.

Thank you.